How-To Library

Informatica

Data Domain Discovery in Test Data Management

© Copyright Informatica LLC 2014, 2021. Informatica Corporation. No part of this document may be reproduced or transmitted in any form, by any means (electronic, photocopying, recording or otherwise) without prior consent of Informatica Corporation. All other company and product names may be trade names or trademarks of their respective owners and/or copyrighted materials of such owners.

Abstract

You can run profiles to discover primary keys, entities, and data domains in Test Data Management (TDM). This article documents the steps to perform data domain discovery in TDM.

Supported Versions

- Test Data Management 9.6.0
- Persistent Data Masking and Data Subset 9.5.2 and HotFixes

Table of Contents

Overview. 2
Profiling in TDM. 3
Example Scenario. 3
Prerequisites. 4
Overview of the Steps. 4
Step 1. Create a Project and Add a Data Source to the Project. 5
Step 2. Create a Data Domain. 6
Step 3. Add the Data Domain to a Policy. 6
Step 4. Add the Policy to the Project. 7
Step 5. Create and Run a Profile. 9
Step 6. Monitor the Profile Job from the Monitor View. 10
Step 7. Review and Apply the Profile Results. 12
Apply the Profile Results to Source Data. 13

Overview

Run a profile in TDM to understand your data before you perform subset or masking operations. Use the results to apply masking rules to multiple columns at a time or instead of manually configuring data subset entities in a subset operation. This document describes how to run a profile on source data to discover data domains. A data domain profile identifies source columns that contain similar data and assigns the same data domain to the columns. A data domain contains a regular expression that defines patterns in the data or patterns in column names. When you run the profile, the profile finds the columns that match the criteria in the regular expression contained in the data domain.

When you configure a profile for data domain discovery, select the tables to search in the data domain discovery operation. Select which data domains to search for in the tables. You can select policies that contain data domains instead of selecting each data domain to search for.

After you run the profile for data discovery, you can view the profile results. The profile results assign source columns to data domains. You can choose which profile results to use when you mask the data in TDM.

Profiling in TDM

Understanding the data before you use it in a masking or subset operation can help you mask similar data with similar rules or discover entities that you can use in a subset operation.

You can use the data discovery feature in TDM to run profiles to understand the data. You can run data domain profiles, entity profiles, and primary key profiles in TDM. You can run profiles only on relational sources. You can run profiles on the following relational source types:

- Oracle
- Microsoft SQL Server
- IBM DB2
- Sybase
- Teradata

Use the ODBC connection type to connect to Sybase and Teradata sources.

You can also create and run profiles in PowerCenter and Informatica Data Quality and export the results. You can then import the profile results into TDM. You can run profiles on relational and nonrelational sources and import the results into TDM. To view the results of a profile on nonrelational sources in the TDM UI, you can run the profile in PowerCenter or Informatica Data Quality and import the results into the TDM repository. TDM supports certain features of profiling in Informatica Data Quality. You can import the results of the profiles that use supported features.

- TDM supports the Enterprise Discovery Profile and Profile options.
- You can import the results of data discovery performed with mapplets created using simple regular expressions. You cannot import results if the mapplets use labeler, tokenizer, content set, or reference tables.
- You can import and view domain discovery results of profiles run in the project. You can import the results of profiles created in folders within the project, but you cannot view the results in TDM.

Tables that you use in a profile must have the same connection as when the table was imported into the repository. If you use a different connection in the profile, you might encounter unexpected results.

You cannot use two tables with the same name in a profile. If a project contains more than one table with the same name, you must run a separate profile for each of the tables.

Example Scenario

Company X provides data analytics services to its customers. It works with large volumes of data, including sensitive data, received from its customers. Company X uses TDM to mask the sensitive data before using the data on its systems.

Applying masking rules to individual columns of data can be tedious, given the volume of data. Multiple columns in different tables might contain data that needs to be masked with the same rules. It would help to be able to apply the masking rules to multiple columns at a time. Because all sensitive data must be masked, it is necessary to identify and mask all sensitive data. Company X uses TDM to run a data domain profile to understand which columns in which tables have similar data and what kind of data. The profile results list columns that belong to a data domain, based on either column title or column content. Company X can analyze the results and apply masking to all columns in a data domain.

In this example, we perform data domain discovery on the following tables:

Table	Columns
ACCOUNT	ACCOUNT_ID ACCOUNT_TYPE BRANCH_ID MIN_BALANCE
CITY	CITY_ID NAME STATE_ID TYPE
BRANCHES	BRANCH_ID CITY_ID
BANKACCOUNTS	ACCOUNT_ID EMP_ID SNO

The objective is to find all columns that have sensitive data like names and account numbers. We can then apply the same masking rules to all columns with names and to all columns with account numbers at the same time. To achieve this, identify masking rules for names and account numbers. Add the rules to a data domain. Add the data domain to a policy in the project. Perform data domain discovery on the data. Analyze the results and directly apply the masking rules in the policy to the columns in the data domain.

Prerequisites

This document describes how to perform data domain discovery on source data in TDM. Before you can perform the tasks described in this document, you must ensure you have all the prerequisites in place.

Before you run a profile in TDM, perform the following tasks:

- Install and configure the compatible version of Informatica services. TDM 9.5.2 and hotfixes work with Informatica services 9.5.1. TDM 9.6.0 works with Informatica services 9.6.0.
- If you have installed Informatica services 9.5.1, you must install EBF 12070.
- Install and configure TDM.
- Ensure that data discovery is enabled in TDM.
- Create the required connections in TDM.

For more information about product requirements and supported platforms, see the Product Availability Matrix on Informatica Network:
https://network.informatica.com/community/informatica-network/product-availability-matrices/overview

Overview of the Steps

To discover data domains in source data, perform the following tasks:

1. Create a project and add a data source to the project.
2. Create a data domain.

3. Add the data domain to a policy.
4. Add the policy to the project.
5. Create and run a profile.
6. Monitor the job from the **Monitor** view.
7. Review and apply the profile results.

Step 1. Create a Project and Add a Data Source to the Project

Create a project PROJECT_PROFILE_DD and add the data sources.

1. In Test Data Manager, click **Projects**.

   A list of projects appears.
2. Click **Actions** > **New**.
3. In the **Create Project** dialog box, enter project properties. The following table describes the project properties:

Option	Description
Name	The name of the project. Enter PROJECT_PROFILE_DD.
Description	The description of the project.
PowerCenter Repository	The name of the PowerCenter repository to store the repository folder.
Folder	The name of the project folder in the repository. Default is the project name. You can choose another folder in the repository.
Owner	The name of the user that owns the folder. The folder owner has all permissions on the folder. The default owner is the user who created the folder. You can select another user as the folder owner.

4. Click **OK**.

   The properties of the project appear in Test Data Manager.
5. Click **Actions** > **Import Metadata**.

   The **Import Metadata** window appears.
6. Choose Datasource Connection and select the database connection from the list. This imports metadata from a database connection.
7. Click **Next**.
8. Select the schema to import. You can filter schemas by schema name.
9. Click **Next**.
10. Select the tables to import. You can filter the tables by data source, table name, or table description.
11. Click **Next**.
12. Choose Import Now. This imports the data source immediately.
13. Click **Finish**.

    You can view the progress of the import job in the **Monitor** view. After the job completes, you can access the imported metadata through the **Data Sources** details view.

Step 2. Create a Data Domain

A data domain is an object that represents the functional meaning of a column based on the name of the column or the data the column contains. When you create a data domain, you create a data expression that describes the data format of the column that you want to mask. You can also create multiple metadata expressions that describe probable column names. After you define a data domain, add masking rules to the data domain.

Identify masking rules for names and account numbers.

1. Click **Policies** from the home page.

   The **Policies** view shows a list of policies, data domains, and rules in the TDM repository.
2. Click **Actions** > **New** > **Data Domain**.
3. Enter the name, sensitivity level, and description for the data domain. Create a data domain DD_Names. Click **Next**.
4. Enter a regular expression to filter columns by data pattern of the column data. Enter the following data pattern: `^(\d){8}$`
5. Click **Next** to add regular expressions that filter columns by column name.

   You can add multiple expressions.
6. Enter regular expressions to filter columns by column name. Enter the following metadata pattern: `(?i)(a(cct|ccount)_?(number|num|nbr|no))`
7. Click **Next** to apply preferred masking rules to the data domain.
8. To add preferred masking rules to the data domain, click **Add Rules**. Add the two rules identified for names and account numbers to the data domain.

   The **Add Rules** dialog box appears.
9. Select the data masking rules to add.
10. Click **OK**.
11. Enable one rule as the default rule.
12. Click **Finish**. The data domain properties page appears.

    View the data and metadata patterns included and the rules included in the data domain.

Step 3. Add the Data Domain to a Policy

You cannot add a data domain directly to a project. Add the data domain to an existing policy that you can add to a project, or create a new policy to add the data domain. Create a new policy Policy_Names and add the data domain created in the previous step to the policy.

1. In the **Policies** view, click **Actions** > **New** > **Policy**.

   The **New Policy** dialog box appears.
2. Enter a name and optional description for the policy and click **Next**. Create a policy Policy_Names.
3. To add data domains to the policy, click **Add Data Domains**.
4. From the list, select the data domain DD_Names.
5. Click **Finish**.

   The policy appears in the **Policies** view.

Step 4. Add the Policy to the Project

To use a policy in a profile, you must add the policy to the project in which you create and run the profile. Add the policy Policy_Names to the project PROJECT_PROFILE_DD.

1. To access the projects, click **Projects**.

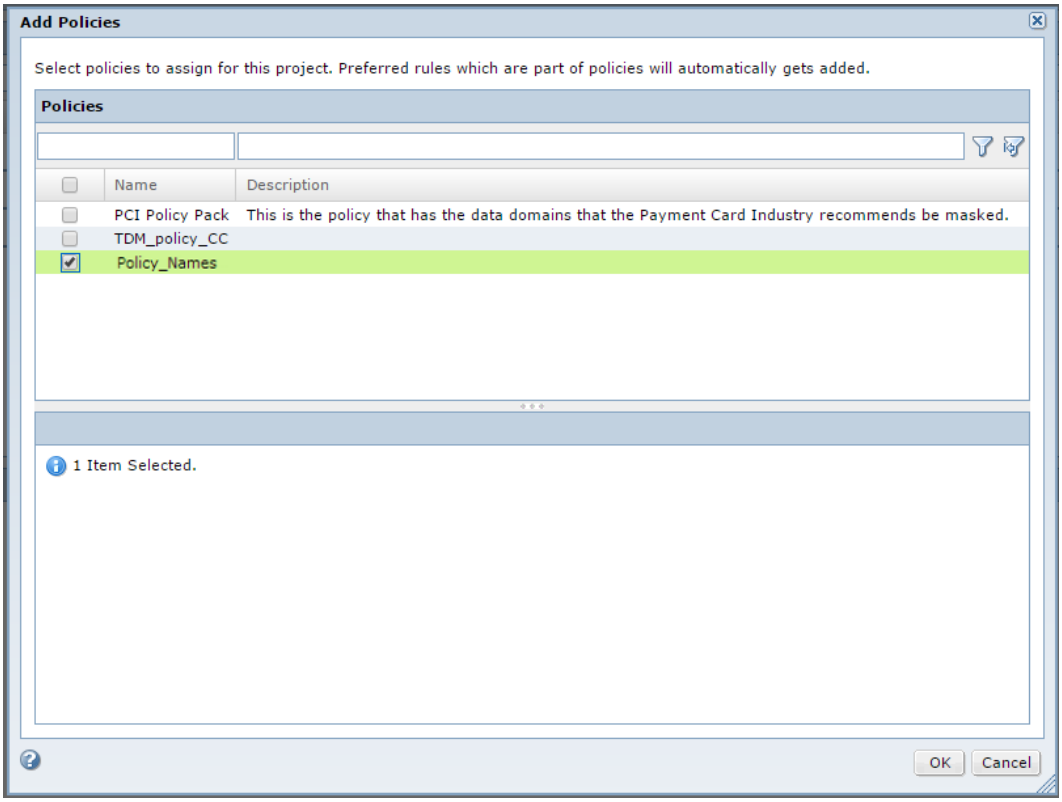
   A list of projects appears.

2. Select the project that you want to edit. Open the project PROJECT_PROFILE_DD.

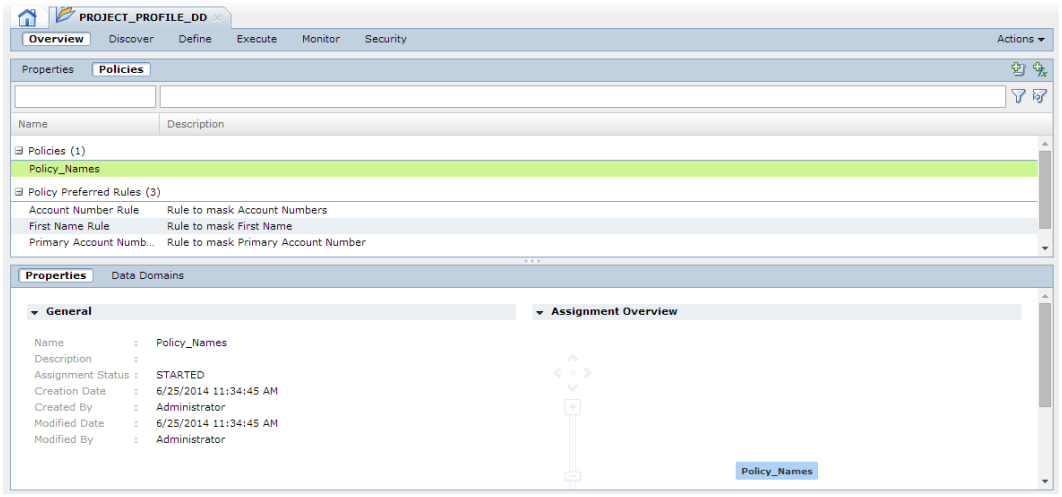
   The project opens in a separate tab with project properties and details about data sources.

3. In the **Policies** view, click **Actions** > **Add Policies**.

4. In the **Add Policies** page, browse to and select the policy, and click **OK**.



The policy is added to the project. You can view the data domains and the preferred rules that are added to the policy.



Step 5. Create and Run a Profile

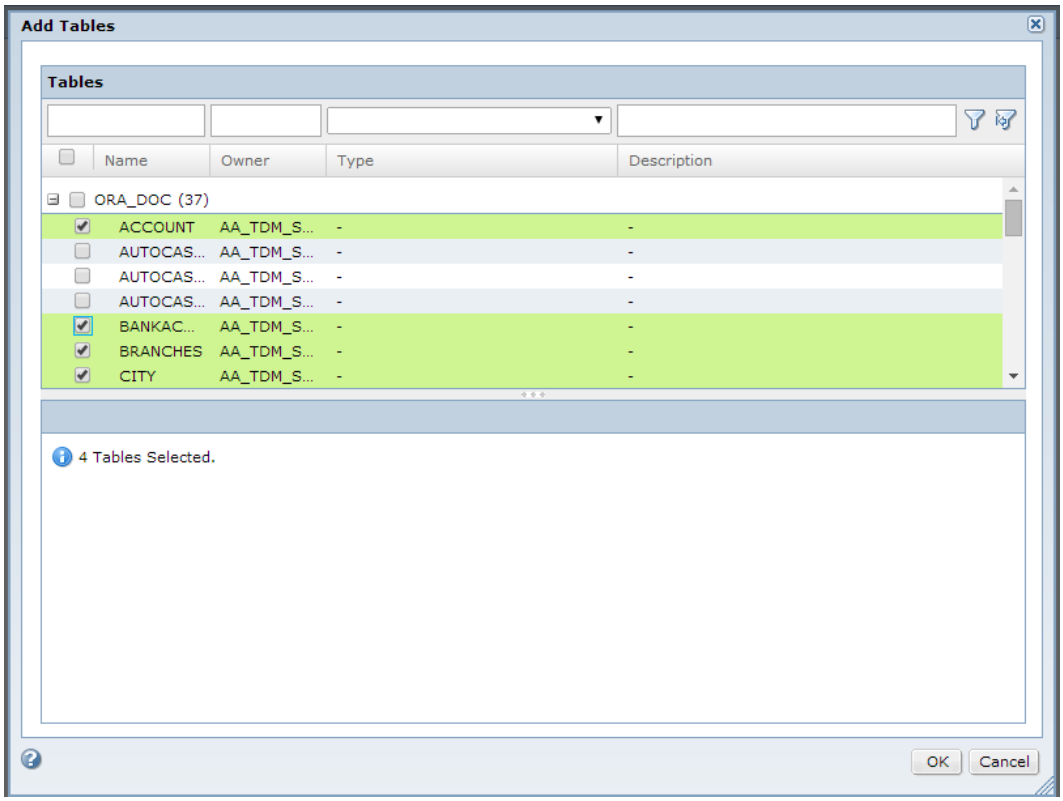
After you create the required data domain and add it to the project, create and run a data domain profile in the project. Create a profile Profile_DD_Names in the project PROJECT_PROFILE_DD.

A project must contain policies before you can create a data domain profile. The policies contain the data domains that you can use in a profile for data discovery. Before you perform this step, ensure that you have added the policy to the project.

Perform data domain discovery on source data to identify data that matches the data format defined in this data domain. Then apply masking rules in the data domain to the columns that match the data domain.

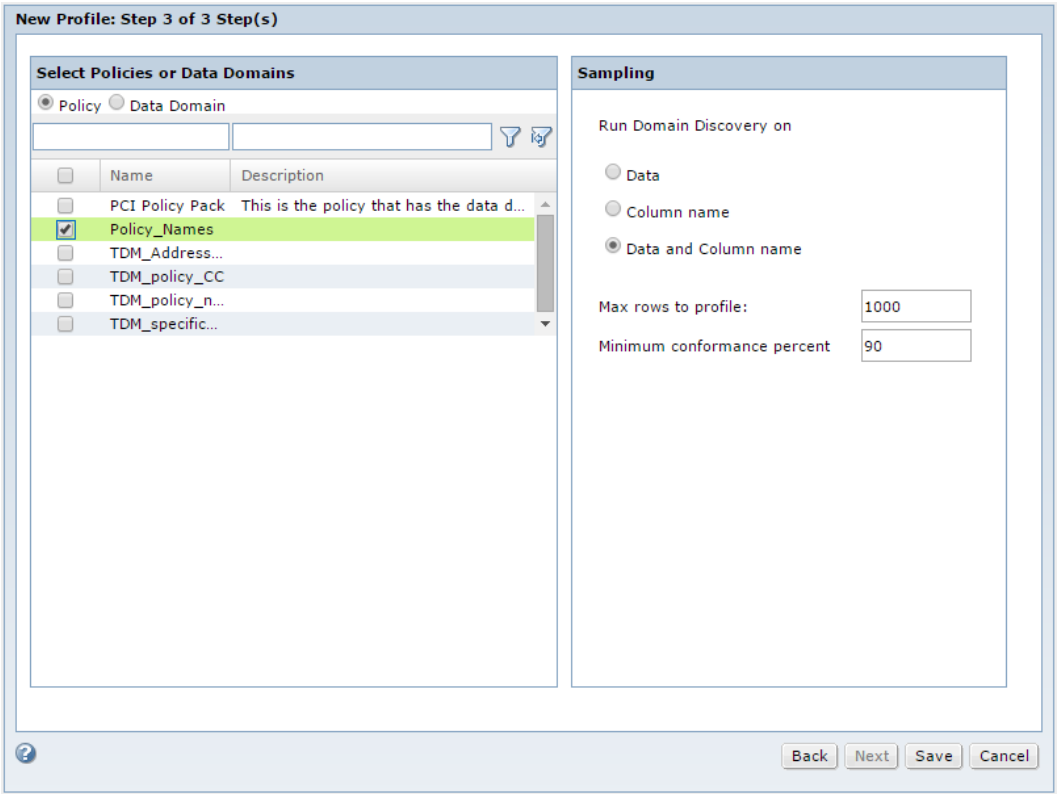
1. Open the project and click the **Discover** view.
2. Click the **Profile** view.

   The **Profile** view shows a list of profiles in the project.
3. Click **Actions** > **New Profile** to create a new profile.
4. In the **New Profile** dialog box, enter the profile name and description. Choose to create a data domain profile. Select the Data Domain check box.
5. Click **Add Tables** to add tables to the profile.
6. Select the tables that you want to add and click **OK**. Select tables ACCOUNT, BANKACCOUNT, BRANCHES, and CITY.
7. Click **Next**.
8. In the **Select Sampling Options** pane, choose whether to add policies or data domains to the profile. When you select a policy, Test Data Manager includes all the data domains in the policy.



Test Data Manager returns a list of policies or data domains in the pane.

9. Select the policies or the data domains to profile. Select the policy Policy_Names.
10. In the **Sampling** panel, select whether to run data discovery on the source data, the column name, or the data and the column name.



You can run a profile for column metadata and then run it again for the source data. Run the profile on the data and column name.

11. Enter the maximum number of rows to include in the profile.
12. Enter the minimum conformance percent.

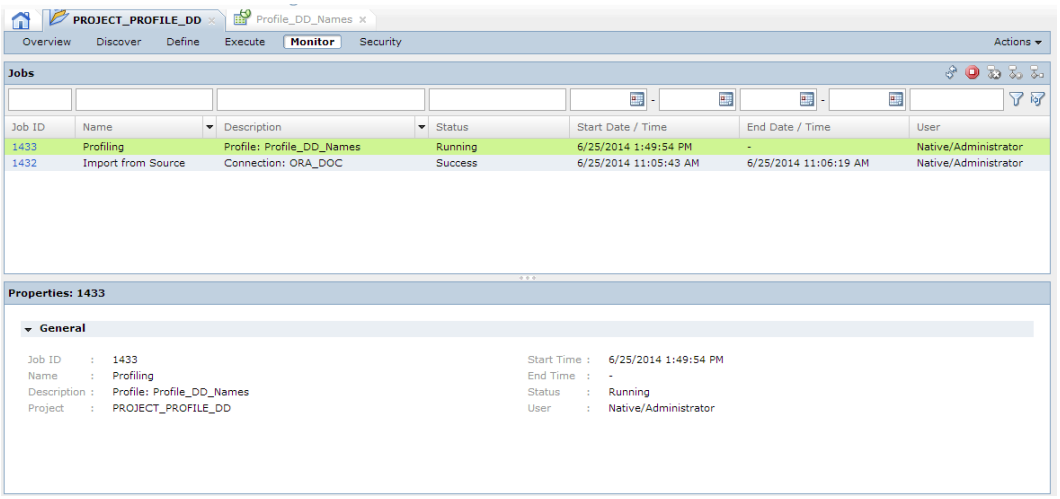
    All rows might not conform to the data domain expression pattern. You can enter a minimum percentage of conformance.
13. Click **Save**.
14. Click **Actions** > **Execute**.
15. Select the connection for the data source. Use the same connection that you used to import the table in the repository.
16. Click **Execute**.

Step 6. Monitor the Profile Job from the Monitor View

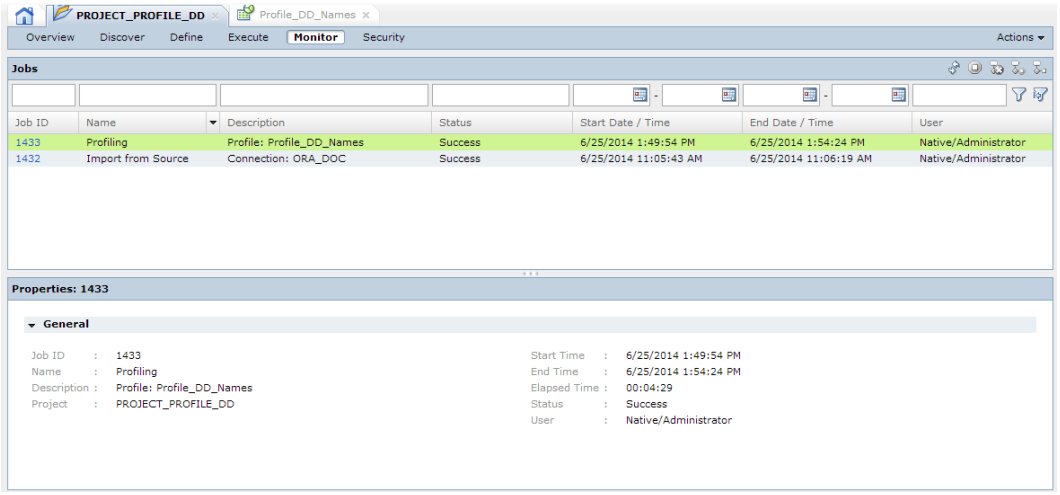
After you run the profile, you can check the status of the profile job.

1. Open the project and click **Monitor**. The list of jobs for the project appear.

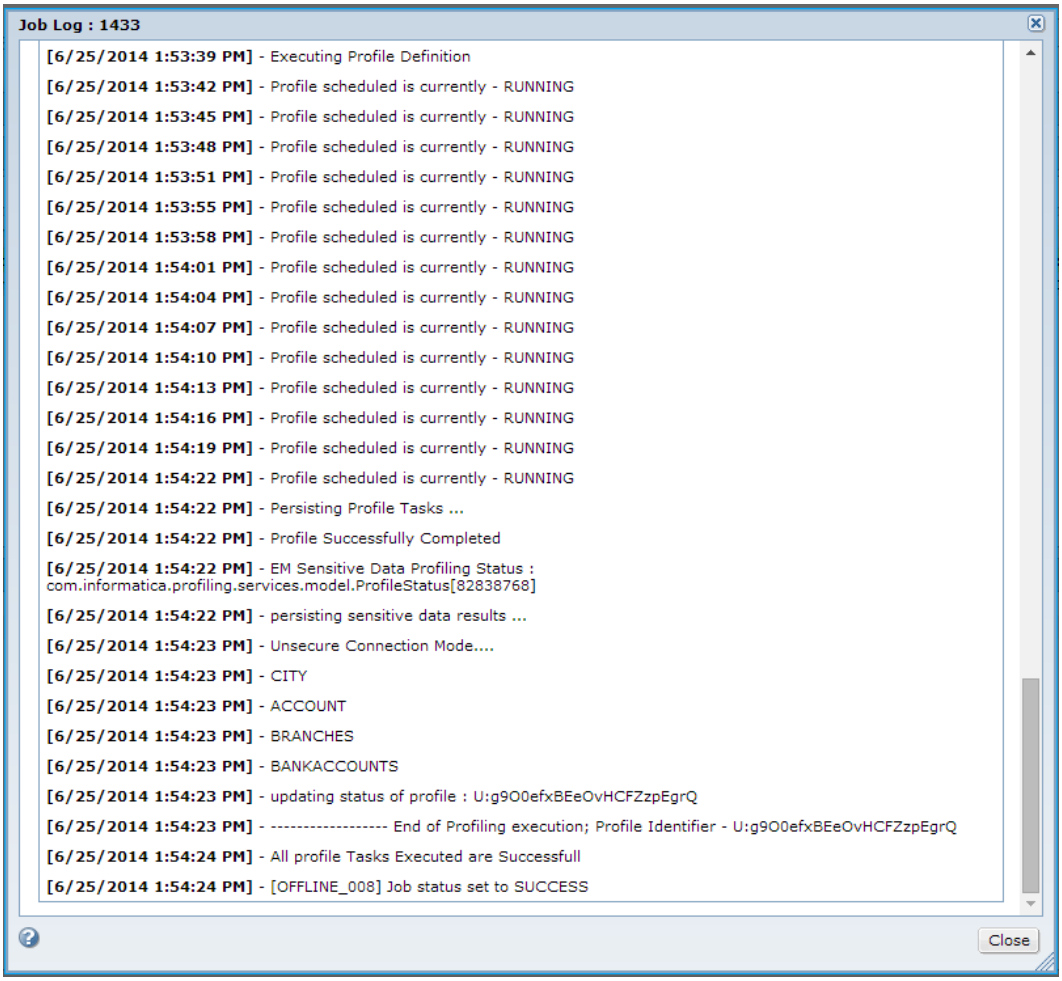
2. Select the job to view the job details in the **Properties** pane.



The status updates when the job finishes.



3. Click the job ID to view the logs on the job log page. The TDM server generates a log file that you can view to debug problems when a TDM job fails.



Step 7. Review and Apply the Profile Results

The data domain profile results show a list of source columns and valid data domains to assign to the columns. You can select which data domain candidates to use for data masking from the profile results.

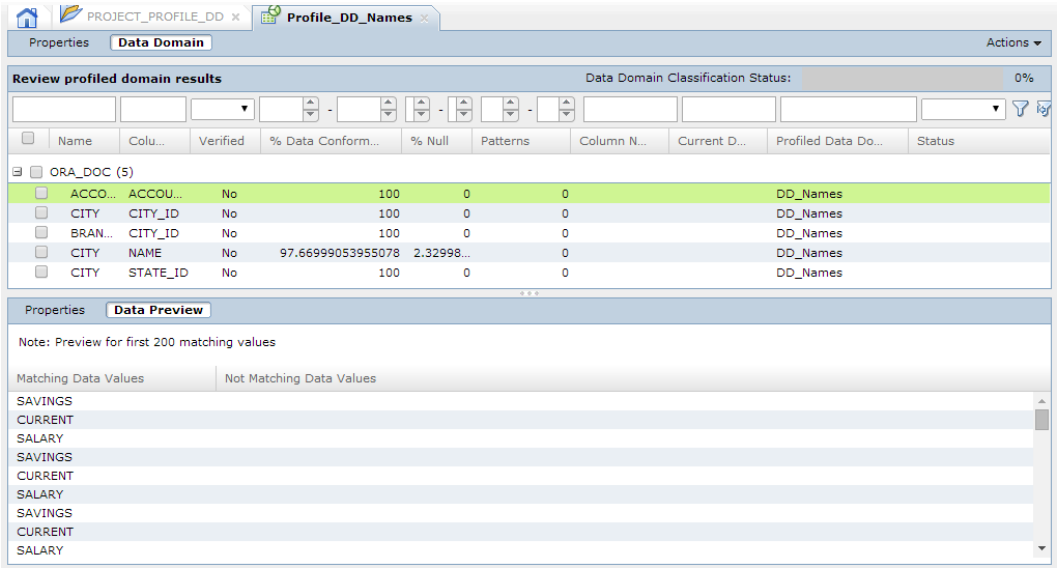
1. Close the profile and open it again.
2. Click the **Data Domain** view.
3. Select a column and click the **Data Preview** tab to view the source data of the selected column.

   The data viewer displays the first 200 records of the columns returned by the data domain profile.
4. Verify the data domain suggested in the **Profiled Data Domain** column in the results.
5. Select **Approve** or **Reject** from the Status column and click **Save** to approve or reject the data domain.
6. Repeat this for all rows.

   You can assign rules in the data domain to each column after you approve the suggested data domain.

7. Mark the data domain classification as completed after you finish approving all the results. Click **Actions** > **Mark domain classification as completed**.

   Completing the data domain classification does not affect any process. Use this method to verify that you reviewed all the results.



Apply the Profile Results to Source Data

You can assign the rules in the data domain to each column after you approve the suggested data domain in the profile results. Assign rules in the **Define | Data Masking** view. The preferred rules for the data domain appear at the top of the list in the Masking Rule column. You can apply the default data domain rules to multiple columns at a time.

▶ In the project, click **Define | Data Masking** to access the **Data Masking** view.

- To assign rules to one column at a time, perform the following steps:
  1. Select a column to assign a masking rule.
  2. If the **Domain** value is blank for the column, click the **Policy** column and choose a policy that contains the data masking rule that you want to assign.
  3. Click inside the **Masking Rule** column to view the list of available rules. The data domain preferred rules appear at the top of the list. The other rules in the policy appear at the bottom of the list.
  4. Select a masking rule.
  5. Click **Save** for each column that you update.
- To assign default data domain rules to multiple columns at the same time, perform the following steps:
  1. Select the columns to which you want to assign default values.
  2. Click **Rule Assignment**. The **Rule Assignments** dialog box appears.
  3. Select the columns to which you want to apply the default values. You can select the check box at the top of the dialog box to select all rows.
  4. Click **Default Assignments**. Test Data Manager updates each column with the default rule.
  5. Click **Save**.

Author

Sadhana Kamath

Acknowledgements

The author would like to acknowledge Praveen Parupudi for his technical assistance.